

Lecture 2: Stationary Time-Series Processes I

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Recap

Last Class

Time Series: Trend, Seasonal, and Stochastic Components

- Time-series data $\{X_t\}_{t=1}^T$ is a collection of observations indexed by time.
 - Time is discrete and equally divided.

- Classical decomposition: $X_t = \underbrace{m_t}_{\text{Trend component}} + \underbrace{s_t}_{\text{Seasonal component}} + \underbrace{\tilde{X}_t}_{\text{Stochastic process}}$

- Our focus: $X_t = \underbrace{\tilde{X}_t}_{\text{Stochastic process}}$

- We drop the **trend** and **seasonal** components because
 - These two components are obvious.
 - These two components make $\{X_t\}_{t=1}^T$ not “well-behaved”.
- Today, we make the meaning of “well-behaved” precise.

Outline

- Preliminary: Math Essentials
- Stationarity
- Three Fundamental Processes for Time-Series Analysis
- Random Walk

Stationary Time-Series Processes

Preliminary: Math Essentials

Math Essentials

- Random variable X (rv X):
 - subject to “fluctuations.”
 - characterized by probability distributions.
- Probability mass/density function (PMF/PDF):
 - X is discrete: $P(X = x)$
 - X is continuous: $f(x)$
- Cumulative distribution function (CDF):
 - X is discrete: $P(X \leq x) = \sum_{u \leq x} P(X = u)$
 - X is continuous: $P(X \leq x) = \int_{-\infty}^x f(u) du =: F(x)$

Math & Statistics Essentials

- Mean of X : $\mu_X := E[X]$
 - X is discrete: $E[X] = \sum_x xP(X = x)$
 - X is continuous: $E[X] = \int_x x dF(x)$
- Expectation $E[\cdot]$ is a linear operator: for rv's X and Y , and a constant $a \in \mathbb{R}$,
 - (1) Additivity: $E[X + Y] = E[X] + E[Y]$
 - (2) Homogeneity: $E[aX] = aE[X]$
- Useful properties:
 - $E[a] = a$
 - $E[aX + bY] = E[aX] + E[bY] = aE[X] + bE[Y]$
 - $E[a_1X_1 + a_2X_2 + \dots + a_NX_N] = a_1E[X_1] + a_2E[X_2] + \dots + a_NE[X_N] = \sum_{i=1}^N a_iE[X_i]$

Math & Statistics Essentials

- Variance of X : $\sigma_X^2 := \text{Var}(X) = E[(X - \mu_X)^2]$
 - $\text{Var}(X) = E[X^2] - \mu_X^2$

Proof.

$$\begin{aligned}\text{Var}(X) &= E[(X - \mu_X)^2] = E[X^2 - 2X\mu_X + \mu_X^2] \\&= E[X^2] - \underbrace{E[2\mu_X X]}_{2\mu_X E[X]} + \underbrace{E[\mu_X^2]}_{\mu_X^2} \\&= E[X^2] - 2\mu_X \underbrace{E[X]}_{\mu_X} + \mu_X^2 \\&= E[X^2] - \mu_X^2\end{aligned}$$

- Standard deviation of X : $\sigma_X := \sqrt{\text{Var}(X)}$

Math & Statistics Essentials

- Covariance of X and Y : $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

- $\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y$

Proof.

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] = E[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y] \\&= E[XY] - \underbrace{E[\mu_Y X]}_{\mu_Y E[X]} - \underbrace{E[\mu_X Y]}_{\mu_X E[Y]} + \underbrace{E[\mu_X \mu_Y]}_{\mu_X \mu_Y} \\&= E[XY] - \mu_Y \underbrace{E[X]}_{\mu_X} - \mu_X \underbrace{E[Y]}_{\mu_Y} + \mu_X \mu_Y \\&= E[XY] - \mu_X \mu_Y\end{aligned}$$

- Correlation of X and Y : $\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

- Standardized between -1 and 1

- Independent of units of measurement

Math & Statistics Essentials

- Variance $\text{Var}(\cdot)$ is **NOT** a linear operator: for rv's X and Y , and a constant $a \in \mathbb{R}$,
 - (1) Non-additive: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) (\neq \text{Var}(X) + \text{Var}(Y))$
 - (2) Non-homogeneous: $\text{Var}(aX) = a^2 \text{Var}(X) (\neq a\text{Var}(X))$
- Useful properties:
 - $\text{Var}(a) = E[(a - E[a])^2] = 0$
 - $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab\text{Cov}(X, Y)$
 - $\text{Var}(a_1X_1 + a_2X_2 + \dots + a_NX_N) = a_1^2 \text{Var}(X_1) + a_2^2 \text{Var}(X_2) + \dots + a_N^2 \text{Var}(X_N)$
 $+ 2a_1a_2 \text{Cov}(X_1, X_2) + \dots + 2a_1a_N \text{Cov}(X_1, X_N)$
 $+ 2a_2a_3 \text{Cov}(X_2, X_3) + \dots + 2a_2a_N \text{Cov}(X_2, X_N)$

 $+ 2a_{N-2}a_{N-1} \text{Cov}(X_{N-2}, X_{N-1}) + 2a_{N-2}a_N \text{Cov}(X_{N-2}, X_N)$
 $+ 2a_{N-1}a_N \text{Cov}(X_{N-1}, X_N)$
 $= \sum_{i=1}^N a_i^2 \text{Var}(X_i) + 2 \sum_{i>j} a_i a_j \text{Cov}(X_i, X_j)$

Math & Statistics Essentials

- Covariance $\text{Cov}(\cdot)$ is a linear operator

(1) Additivity: $\text{Cov}(X + Z, Y) = \text{Cov}(X, Y) + \text{Cov}(Z, Y)$

$$\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$$

(2) Homogeneity: $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$

$$\text{Cov}(X, bY) = b\text{Cov}(X, Y)$$

- Useful properties:

– $\text{Cov}(a, Y) = 0$

– $\text{Cov}(a, b) = 0$

– $\text{Cov}(X, X) = \text{Var}(X)$

– $\text{Cov}(aX + cW, bY + dZ) = ab\text{Cov}(X, Y) + ad\text{Cov}(X, Z) + bc\text{Cov}(W, Y) + cd\text{Cov}(W, Z)$

– $\text{Cov}(\sum_{i=1}^N a_i X_i, \sum_{j=1}^M b_j Y_j) = \sum_{n=1}^N \sum_{j=1}^M a_i b_j \text{Cov}(X_i, Y_j)$

- Two independent random variables X, Y :
 - Definition: $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$
 - Implication 1: $E[XY] = E[X]E[Y]$
 - Implication 2: $Cov(X, Y) = 0$

Math & Statistics Essentials

- Two independent random variables X, Y :
 - Definition: $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$
 - Implication 1: $E[XY] = E[X]E[Y]$

Proof: Assuming that X and Y are discrete independent random variables,

$$\begin{aligned} E[XY] &= \sum_x \sum_y xy P(X \leq x, Y \leq y) \\ &= \sum_x \sum_y xy P(X \leq x)P(Y \leq y) \\ &= \underbrace{\sum_x xP(X \leq x)}_{E[X]} \underbrace{\sum_y yP(Y \leq y)}_{E[Y]} \\ &= E[X]E[Y] \end{aligned}$$

- Implication 2: $\text{Cov}(X, Y) = 0$

- Two independent random variables X, Y :
 - Definition: $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$
 - Implication 1: $E[XY] = E[X]E[Y]$
 - Implication 2: $Cov(X, Y) = 0$

Proof: Assuming that X and Y are independent random variables,

$$\begin{aligned}Cov(X, Y) &= E[XY] - E[X]E[Y] \\&= \underbrace{E[XY]}_{E[X]E[Y]} - E[X]E[Y] \\&= 0\end{aligned}$$

Stationary Time-Series Processes

Introduction

Time-Series Processes

Definition (Time-Series Processes)

A time-series process $\{X_t\}_{t=1}^T$ is characterized by its joint probability distribution:

$$P(X_1 \leq x_1, \dots, X_T \leq x_T).$$

- $P(X_1 \leq x_1, \dots, X_T \leq x_T)$: the joint cumulative distribution function (joint CDF)
- Time-series analysis essentially boils down to analyzing the joint CDF.

Stationary Time-Series Processes

Stationarity

Strong Stationarity

Definition (Strongly Stationary)

A time-series process $\{X_t\}_{t=-\infty}^{\infty}$ is strongly stationary if for any $k \in \mathbb{N}$,

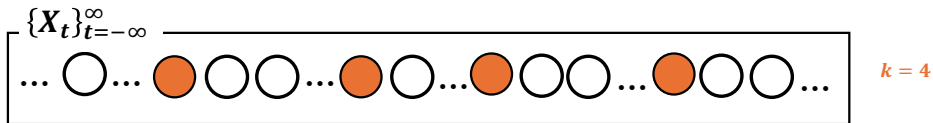
$$P(X_{t_1} \leq x_{t_1}, \dots, X_{t_k} \leq x_{t_k}) = P(X_{t_1+h} \leq x_{t_1}, \dots, X_{t_k+h} \leq x_{t_k}) \text{ for all } h \geq 1.$$

- For any finite subset of $\{X_t\}_{t=-\infty}^{\infty}$, the joint probability distribution of $\{X_t\}_{t=t_1}^{t_k}$ is the same as the joint probability distribution of $\{X_t\}_{t=t_1+h}^{t_k+h}$, whatever $h \geq 1$ is.
 $\implies$ Shifting the time index does not affect the joint probability distribution.
- e.g., a process of independent and identically distributed random variables
- Strong stationarity pertains to the entire shape of the joint probability distributions.
 - Very rare.

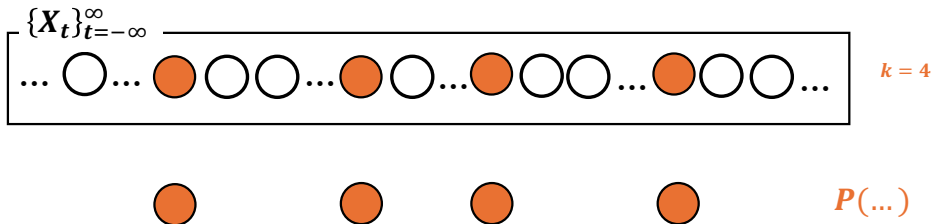
Strong Stationarity



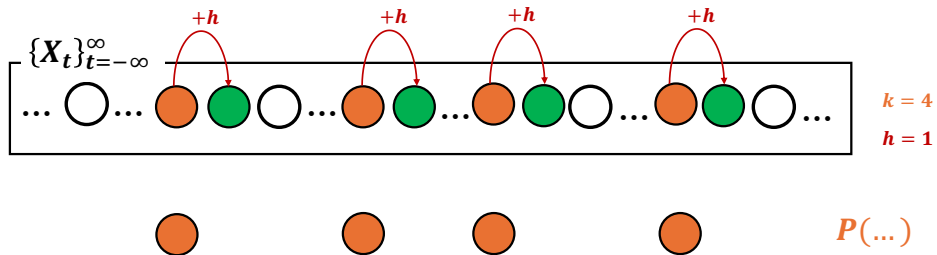
Strong Stationarity



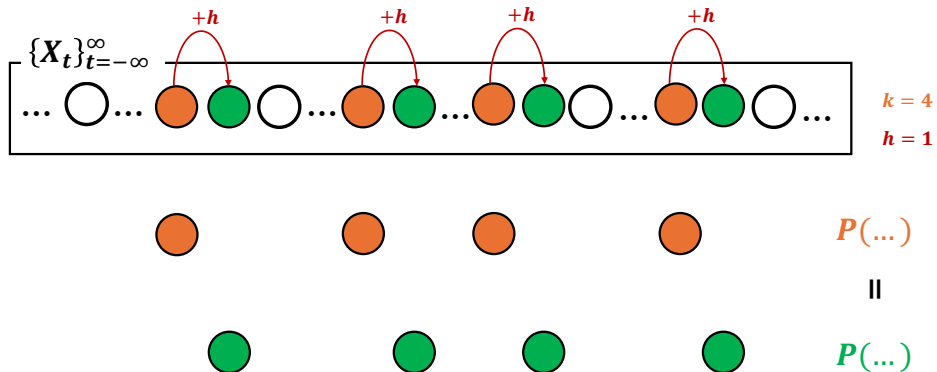
Strong Stationarity



Strong Stationarity



Strong Stationarity



Strong Stationarity

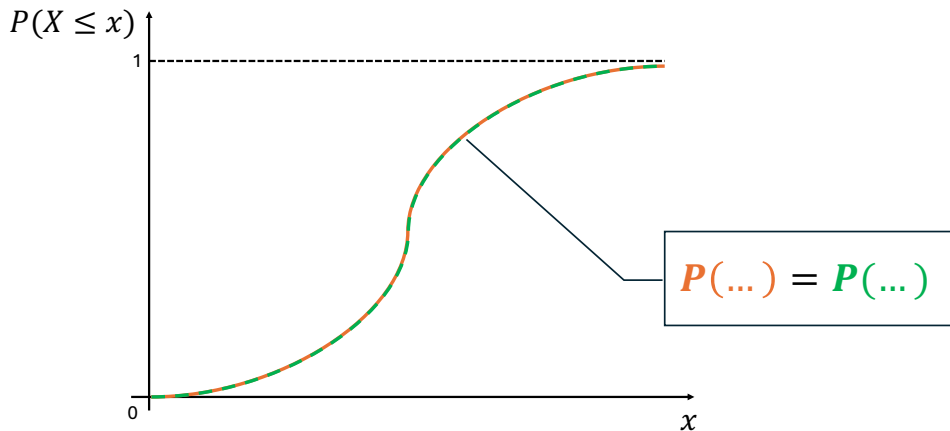


Figure 1: Probability distribution function of X

Moments Characterizing a Time Series

- **Idea:** To focus on the **three key aspects** of the joint probability distribution.
- Suppose that $\{X_t\}$ is a time series

(1) **Mean:**

$$\mu_X(t) := E[X_t]$$

(2) **Variance:**

$$\sigma_X^2(t) := \text{Var}(X_t) = E[(X_t - \mu_t)^2]$$

(3) **Autocovariance:** for $h = 1, 2, 3, \dots, t-1$

$$\gamma_{X,h}(t) := \text{Cov}(X_t, X_{t-h}) = E[(X_t - \mu_{X,t})(X_{t-h} - \mu_{X,t-h})]$$

– Notice that: $\sigma_X^2 = \gamma_{X,0}$

(3)' **Autocorrelation:** for $h = 1, 2, 3, \dots, t-1$

$$\rho_{X,h}(t) := \text{Corr}(X_t, X_{t-h}) = \frac{\gamma_{X,h}(t)}{\sigma_X(t)\sigma_X(t-h)}$$

Weak Stationarity

Definition (Weakly Stationary)

A time-series process $\{X_t\}_{t=-\infty}^{\infty}$ is weakly (covariance) stationary if its mean, variance, and autocovariances are constant across time; that is,

$$(1) \mu_X(t) = \mu_X$$

$$(2) \sigma_X^2(t) = \sigma_X^2$$

$$(3) \gamma_{X,h}(t) = \gamma_{X,h} \ (\Longleftrightarrow \ \rho_{X,h}(t) = \rho_{X,h})$$

- The first- and second-order moments of the joint probability distributions are independent of time t .
 - $\gamma_{X,h}(t)$ (and $\rho_{X,h}(t)$) can still depend on the lag h .
 - Autocorrelation function (ACF): a plot of $\rho_{X,h}(t)$ with respect to the lag h .

Weak Stationarity

- Visually, a flat-looking series without trend or seasonal components.

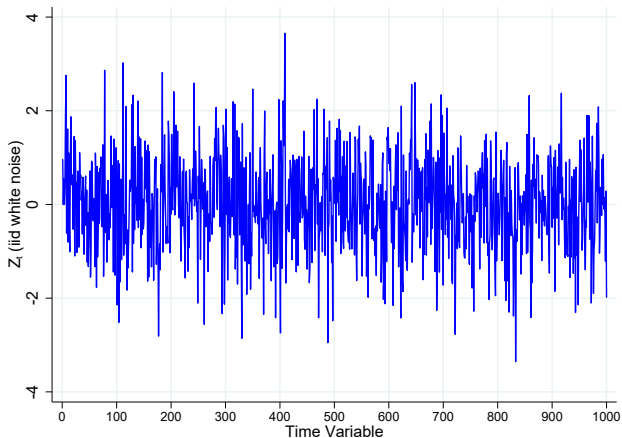


Figure 2: A time-series plot of a weak stationary process

Stationary Time-Series Processes

Three Fundamental Processes in Time Series

White Noise Process

(1) **White noise:** $X_t \sim WN(0, \sigma_X^2)$

- $E[X_t] = 0$, $Var(X_t) = \sigma_X^2$, and $Cov(X_t, X_{t+h}) = 0$ for all t, h .
- $P(\cdot)$ is unspecified.

(2) **Independent and Identically Distributed (iid) White noise:** $X_t \sim IWN(0, \sigma_X^2)$

- $E[X_t] = 0$, $Var(X_t) = \sigma_X^2$, and $Cov(X_t, X_{t+h}) = 0$ for all t, h .
- Independent: $P(X_{t_1} \leq x_{t_1}, \dots, X_{t_k} \leq x_{t_k}) = P(X_{t_1} \leq x_{t_1}) \cdots P(X_{t_k} \leq x_{t_k})$.
- Identically distributed: $P(X_{t_1} \leq \alpha) = P(X_{t_2} \leq \alpha) = \dots = P(X_{t_k} \leq \alpha)$ for any $\alpha \in \mathbb{R}$.

(3) **Gaussian White noise:** $X_t \sim GWN(0, \sigma_X^2)$

- $E[X_t] = 0$, $Var(X_t) = \sigma_X^2$, and $Cov(X_t, X_{t+h}) = 0$ for all t, h .
- $P(\cdot)$ is a Gaussian distribution (with zero covariance).
 $\implies \{X_t\}$ are iid, i.e., $\{X_t\}$ is a special case of $IWN(0, \sigma_X^2)$.

White Noise Process

(1) **White noise**: $X_t \sim WN(0, \sigma_X^2)$

- $E[X_t] = 0$, $Var(X_t) = \sigma_X^2$, and $Cov(X_t, X_{t+h}) = 0$ for all t, h .
- $P(\cdot)$ is unspecified.

Q1. Is **White noise weakly stationary**?

$\implies$ Yes!

Q2. Is **White noise strongly Stationary**?

$\implies$ Not necessarily...

White Noise Process

(2) **IID White noise**: $X_t \sim IWN(0, \sigma_X^2)$

- $E[X_t] = 0$, $Var(X_t) = \sigma_X^2$, and $Cov(X_t, X_{t+h}) = 0$ for all t, h .
- Independent: $P(X_{t_1} \leq x_{t_1}, \dots, X_{t_k} \leq x_{t_k}) = P(X_{t_1} \leq x_{t_1}) \cdots P(X_{t_k} \leq x_{t_k})$.
- Identically distributed: $P(X_{t_1} \leq \alpha) = P(X_{t_2} \leq \alpha) = \dots = P(X_{t_k} \leq \alpha)$ for any $\alpha \in \mathbb{R}$.

Q1. Is **IID White noise weakly stationary**?

$\implies$ Yes!

Q2. Is **IID White noise strongly stationary**?

$\implies$ Yes!

White Noise Process

(2) **IID White noise**: $X_t \sim IWN(0, \sigma_X^2)$

Q2. Is **IID White noise strongly stationary**? $\rightarrow$ Yes!

By the independence of $\{X_t\}_{t=t_1}^{t_k}$,

$$P(X_{t_1} \leq x_{t_1}, X_{t_2} \leq x_{t_2}, \dots, X_{t_k} \leq x_{t_k}) = P(X_{t_1} \leq x_{t_1})P(X_{t_2} \leq x_{t_2}) \dots P(X_{t_k} \leq x_{t_k}).$$

By the independence and identically distributed nature of $\{X_t\}_{t=t_1+h}^{t_k+h}$,

$$\begin{aligned} P(X_{t_1+h} \leq x_{t_1}, X_{t_2+h} \leq x_{t_2}, \dots, X_{t_k+h} \leq x_{t_k}) &= \underbrace{P(X_{t_1+h} \leq x_{t_1})}_{P(X_{t_1} \leq x_{t_1})} \underbrace{P(X_{t_2+h} \leq x_{t_2})}_{P(X_{t_2} \leq x_{t_2})} \dots \underbrace{P(X_{t_k+h} \leq x_{t_k})}_{P(X_{t_k} \leq x_{t_k})} \\ &= P(X_{t_1} \leq x_{t_1})P(X_{t_2} \leq x_{t_2}) \dots P(X_{t_k} \leq x_{t_k}). \end{aligned}$$

Hence, for any k and h ,

$$\begin{aligned} P(X_{t_1} \leq x_{t_1}, X_{t_2} \leq x_{t_2}, \dots, X_{t_k} \leq x_{t_k}) &= P(X_{t_1} \leq x_{t_1})P(X_{t_2} \leq x_{t_2}) \dots P(X_{t_k} \leq x_{t_k}) \\ &= P(X_{t_1+h} \leq x_{t_1}, X_{t_2+h} \leq x_{t_2}, \dots, X_{t_k+h} \leq x_{t_k}) \end{aligned}$$

White Noise Process

(3) **Gaussian White noise**: $X_t \sim GWN(0, \sigma_X^2)$

- $E[X_t] = 0$, $Var(X_t) = \sigma_X^2$, and $Cov(X_t, X_{t+h}) = 0$ for all t, h .
- $P(\cdot)$ is a Gaussian distribution.

Q1. Is **Gaussian White noise** **weakly stationary**?

$\implies$ Yes!

Q2. Is **Gaussian White noise** **strongly stationary**?

$\implies$ Yes!

Stationary Time-Series Processes

Random Walk

Random Walk

- **Random walk:** $\{X_t\}$ is a random walk if

$$X_t = X_{t-1} + \varepsilon_t,$$

where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.

- By repeated substitution,

$$X_t = X_0 + \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t \text{ with } \{\varepsilon_s\}_{s=1}^t \sim WN(0, \sigma_\varepsilon^2),$$

- Accumulated path of white noises over time.
- In what follows, $X_0 = 0$; that is,

$$X_t = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t \text{ with } \{\varepsilon_s\}_{s=1}^t \sim WN(0, \sigma_\varepsilon^2)$$

Random Walk

- A random walk exhibits **strong dependence over time**, so **long-memory process**



Figure 3: Simulated Random Walk

Random Walk

- Random walk:

$$X_t = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t \text{ with } \{\varepsilon_s\}_{s=1}^t \sim WN(0, \sigma_\varepsilon^2)$$

- Q. Is a random walk strongly stationary and/or weakly stationary?
- Q. What do the three key moments (i.e., mean, variance, autocovariance) look like?

Mean:

$$\begin{aligned} E[X_t] &= E[\varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t] \\ &= \underbrace{E[\varepsilon_1]}_0 + \underbrace{E[\varepsilon_2]}_0 + \dots + \underbrace{E[\varepsilon_t]}_0 \\ &= 0 \end{aligned}$$

Random Walk

- Random walk:

$$X_t = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t \text{ with } \{\varepsilon_s\}_{s=1}^t \sim WN(0, \sigma_\varepsilon^2)$$

Variance:

$$\begin{aligned} \text{Var}(X_t) &= \text{Var}(\varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t) \\ &= \underbrace{\text{Var}(\varepsilon_1)}_{\sigma_\varepsilon^2} + \underbrace{\text{Var}(\varepsilon_2)}_{\sigma_\varepsilon^2} + \dots + \underbrace{\text{Var}(\varepsilon_t)}_{\sigma_\varepsilon^2} \\ &\quad + 2 \underbrace{\text{Cov}(\varepsilon_1, \varepsilon_2)}_0 + 2 \underbrace{\text{Cov}(\varepsilon_1, \varepsilon_3)}_0 + \dots + 2 \underbrace{\text{Cov}(\varepsilon_t, \varepsilon_{t-1})}_0 \\ &= t\sigma_\varepsilon^2 \end{aligned}$$

Random Walk

- Random walk:

$$X_t = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t \text{ with } \{\varepsilon_s\}_{s=1}^t \sim WN(0, \sigma_\varepsilon^2)$$

Autocovariance:

$$\begin{aligned} \text{Cov}(X_t, X_{t-h}) &= \text{Cov}(\varepsilon_1 + \dots + \varepsilon_t, \varepsilon_1 + \dots + \varepsilon_{t-h}) \\ &= \underbrace{\text{Cov}(\varepsilon_1, \varepsilon_1)}_{\sigma_\varepsilon^2} + \underbrace{\text{Cov}(\varepsilon_2, \varepsilon_2)}_{\sigma_\varepsilon^2} + \dots + \underbrace{\text{Cov}(\varepsilon_{t-h}, \varepsilon_{t-h})}_{\sigma_\varepsilon^2} \\ &\quad + \underbrace{\text{Cov}(\varepsilon_1, \varepsilon_2)}_0 + \underbrace{\text{Cov}(\varepsilon_1, \varepsilon_3)}_0 + \dots + \underbrace{\text{Cov}(\varepsilon_t, \varepsilon_{t-h})}_0 \\ &= (t - h)\sigma_\varepsilon^2 \end{aligned}$$

Random Walk

- Random walk:

$$X_t = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t \text{ with } \{\varepsilon_s\}_{s=1}^t \sim WN(0, \sigma_\varepsilon^2)$$

Autocorrelation:

$$\begin{aligned} \text{Corr}(X_t, X_{t-h}) &= \frac{\text{Cov}(X_t, X_{t-h})}{\sqrt{\text{Var}(X_t)}\sqrt{\text{Var}(X_{t-h})}} \\ &= \frac{(t-h)\sigma_\varepsilon^2}{\sqrt{t\sigma_\varepsilon^2}\sqrt{(t-h)\sigma_\varepsilon^2}} = \frac{(t-h)\sigma_\varepsilon^2}{\sqrt{t}\sqrt{t-h}\sigma_\varepsilon^2} \\ &= \sqrt{\frac{t-h}{t}} \end{aligned}$$

Random Walk

- Random walk:

$$X_t = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t \text{ with } \{\varepsilon_s\}_{s=1}^t \sim WN(0, \sigma_\varepsilon^2)$$

- Mean: $E[X_t] = 0$
- Variance: $Var(X_t) = t\sigma_\varepsilon^2$
- Autocovariance: $Cov(X_t, X_{t-h}) = (t-h)\sigma_\varepsilon^2$
- Autocorrelation: $Corr(X_t, X_{t-h}) = \sqrt{\frac{t-h}{t}}$

Q1. Is a random walk weakly stationary?

$\implies$ No...

Q2. Is a random walk strongly stationary?

$\implies$ No...

Random Walk

- A random walk is not weakly stationary.
- But, what about the **first difference** of a random walk?

$$\begin{aligned} X_t &= X_{t-1} + \varepsilon_t \\ \iff \underbrace{X_t - X_{t-1}}_{\Delta X_t} &= \varepsilon_t \end{aligned}$$

- That is,

$$\Delta X_t = \varepsilon_t \sim \text{WN}(\mathbf{0}, \sigma_\varepsilon^2)$$

- **The first difference** of **a random walk** is **weakly stationary**.

Summary & Preview

Summary

- Strong stationarity and weak stationarity
- Three building blocks:
 - WN, IWN, GWN
- Random walk

Preview

- We will study more time-series Processes:
 - MA, AR, ARMA Processes
- We will repeat the same manipulations (e.g., mean, variance, autocovariance)!